TEAM MODEST

ONECASE

A connected workspace for institution-related complaints

Coordinate ownership, proceedings, evidence and outcomes across AICTE bureaus and cells.

Complaint intake | **Accountable ownership** | **Proceeding records** | **Semantic overlap review** | **Deadline monitoring** | **Evidence & closure PDFs**

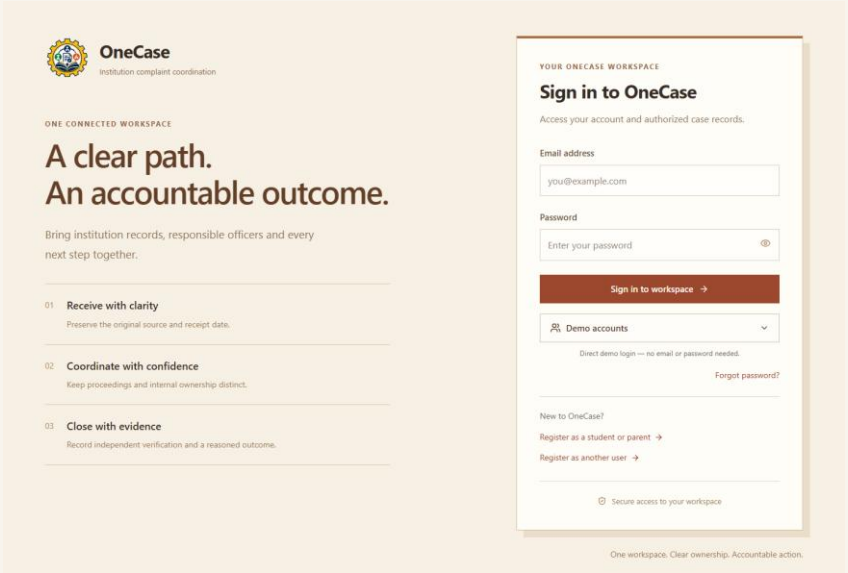
COMPLAINT JOURNEY

Register → assign → coordinate → verify → communicate → close

WHO USES IT

Students, parents, faculty, staff and other stakeholders; approved officers, leads, reviewers and appointed participants.

Original receipt, source, ownership and action history stay connected.



Live: sih-solution-kappa.vercel.app

TECHNICAL APPROACH

FRONTEND · VERCEL
React + TypeScript + Vite

API · RENDER
Flask + SQLAlchemy + Waitress

PERSISTENCE · NEON
PostgreSQL + private S3 bucket

BACKGROUND WORKER
Local models + reminders

SUPPORTING SERVICES
Redis limits; SendGrid email

Browser → Vercel proxy → Render API

LOCAL AI: DESCRIPTION-LEVEL OVERLAP REVIEW

Normalize	MiniLM embeddings	Retrieve top 20	Cross-encoder rerank	Human review

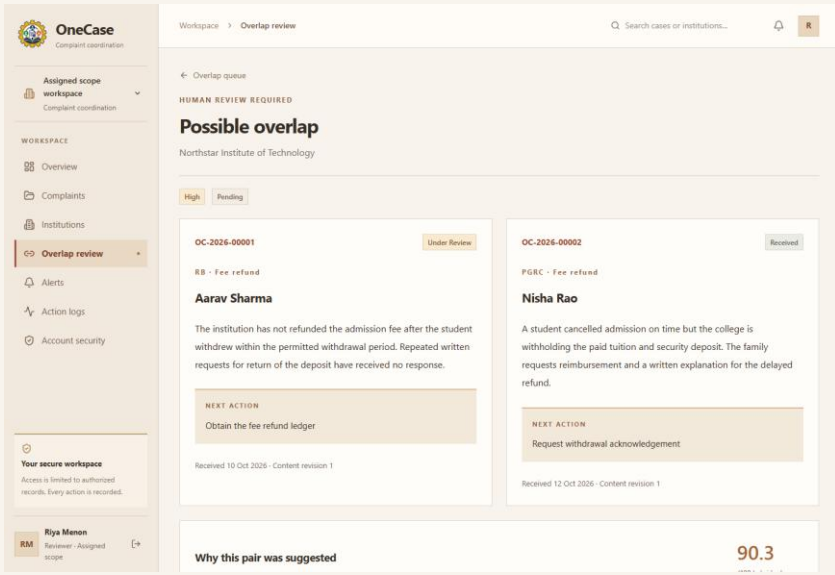
Bounded candidates

Match within the same verified institution. Original complaints stay separate.

Signals, not decisions

Similarity, reranker output and category support triage; scores are not probabilities.

Pinned models run locally. Revision changes invalidate stale decisions.



Actual model output; synthetic complaints

API and worker share database/private storage; service credentials stay server-side.

FEASIBILITY AND VIABILITY

IMPLEMENTED WORKFLOWS

Public intake, ownership/transfers, action history, deadline policies, proceeding records, institution review and closure reports.

DEPLOYMENT REALITY

Vercel frontend; Render API and separate worker. Free API can sleep; the wake flow waits up to 90 seconds. Worker/model compute needs separate resources.

VALIDATION STILL REQUIRED

Domain accuracy, multilingual performance, load/uptime benchmarks and live configuration. No measured savings or certification is claimed.

The wake flow runs on visits; it does not make Render Free an always-on service.

SECURITY IN THE CODE

Authentication

Argon2, hashed sessions, MFA and one-time recovery codes.

Authorization

Case/document/field scope, expiring grants and conflict denial.

Request protection

CSRF, shared proxy secret and rate limits.

Private evidence

Neon storage, SHA-256 checksums, file limits and bounded parsing.

Antivirus bypass active

Unscanned files; parser and access checks remain. Malware protection is reduced.

IMPACT AND BENEFITS

Expected benefits: clearer ownership, traceable decisions and fewer disconnected case records

Complainants

One reference, authorized tracking and closure PDFs.

Officers and bureau leads

Owners, transfers, next actions and deadline reminders.

Reviewers and committees

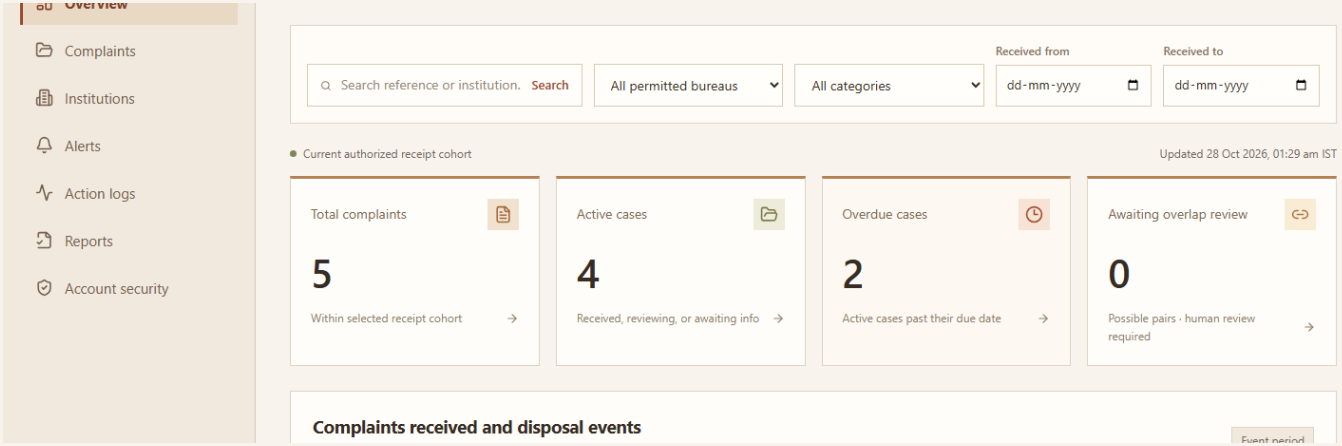
Related-case suggestions and bounded proceeding access.

Administrators and monitoring roles

Approved access, institution review, scoped totals and CSV reports.

PROCEEDINGS AND VERIFIED CLOSURE

Route authority → appoint → record milestones → independently verify compliance → record dispatch/acknowledgement → communicate outcome → freeze PDF



Implemented: access administration, scoped reports and browser install metadata.

Next: bulk CSV import, official registry integration, issue-level matching and domain/load benchmarks.

RESEARCH AND REFERENCES

Research source	Relevant finding	OneCase implementation
[1] AICTE Handbook 2024-2027	SGRC and Ombudsperson responsibilities are defined through grievance procedures.	Proceeding routes, authority records and milestone snapshots; decisions stay with competent authorities.
[2] Sentence-BERT Reimers & Gurevych, 2019	Embeddings enable semantic comparison using cosine similarity.	MiniLM retrieves complaint candidates within the same verified institution.
[3] Sentence Transformers Retrieve & Re-Rank	Retrieve candidates, then rerank the shortlist with a cross-encoder.	Top-20 retrieval and reranking support human review. Scores remain unvalidated.
[4] OWASP Authorization Cheat Sheet	Deny by default; check permissions for every request.	Case/document scope, expiring grants, field redaction and access-denial tests.

[1] AICTE handbook

[2] Sentence-BERT paper

[3] Retrieve & Re-Rank

[4] OWASP authorization